What an Auxiliary Cue Must Carry: LiDAR Body Pose for Pedestrian Trajectory Forecasting

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***Abstract*—Body pose appears to carry pedestrian intent that position history lacks. On the Waymo Open Motion Dataset (WOMD), with SMPL poses from LiDAR, we compare own pose with no pose and with a stranger's pose. On WOMD's tracks, offboard auto-labels refined with future frames, own pose costs 1.7% against no pose at our largest scale ($p<0.001$), as a stranger's does (difference $+0.04\%$, within $\pm 1.7\%$). Under a simulated onboard tracker (30 cm noise) it gains 4.1% ($p=0.004$), and 5.3% over a stranger's ($p=0.002$), but three quarters of the latter is the offboard label returning through the pose pipeline's track fit. Poses taken before that fit gain nothing over no pose (0.4% worse, $p=0.54$). A gain on pedestrians with at least 200 returns (2.0%, $p=0.041$, uncorrected) is absent at the final epoch on held-out scenes (0.9% worse, $p=0.67$). There own pose leads a stranger's by 2.1% ($p=0.12$), rising with the tracker's velocity error from 0.0% to 4.9% across quartiles ($p<0.001$). Our contribution is this leak and the three controls behind these results: a stranger's pose, pre-fit poses, and an oracle that the same branch converts.**

***Index Terms*—pedestrian trajectory prediction, human pose estimation, LiDAR, autonomous driving, offboard labels**

I. Introduction

Predicting where a pedestrian will go is a safety-critical function of an automated vehicle, and body pose is an obvious candidate input. A person's stance, gait phase and facing direction plausibly reveal intent before their position changes [1], [2], and modern perception stacks already produce human mesh or keypoint estimates. Several recent systems condition trajectory forecasts on pose [3]–[6].

We ask first whether a forecaster given pose beats the same forecaster without it, and second whether a difference is due to the pedestrian's own body, which a control fed a stranger's pose decides. On the labelled tracks of the Waymo Open Motion Dataset (WOMD) pose beats neither no pose nor a stranger's pose, and the evidence points to the track it is added to. WOMD's boxes are offboard auto-labels, refined with the whole track, future frames included [7], [8], and against that track the body restates what the model already knows. Against the history a causal onboard tracker would report, own pose beats both, but most of that gain is the offboard label returning through the pose pipeline, which fits the gait to the track. Poses taken before that fit do not beat no pose on held-out scenes, and beat a stranger's only where the body is resolved and the tracker is wrong (Fig. 1).

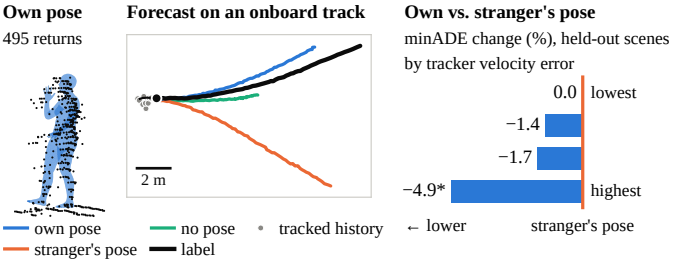


Fig. 1. Under a simulated onboard tracker (30 cm noise), a pedestrian's own SMPL pose from LiDAR lowers minADE against a stranger's pose where the tracker is wrong, by 4.9% in the highest quartile of its velocity error. Bars are $13{,}350$ held-out pedestrians with at least 200 returns and 1 s of future, scored at the final epoch. Against no pose they gain nothing ($+0.9\%$, n.s.). The poses are taken before the track fit, the asterisk marks $p<0.05$, uncorrected, and the example is from Fig. 4, selected on the effect, each model's median seed and best of six modes.

Our contributions are threefold. First, three controls: a stranger's pose, which holds the added branch fixed, poses taken before the pose pipeline fits them to the track, and an oracle through the same branch, which improves minADE by 34.8% and is converted even as a coarse future direction. Second, a label leak they expose. Under a simulated causal tracker own pose beats no pose by 4.1% and a stranger's by 5.3%, but three quarters of the latter is the offboard label returning through the pose pipeline. Third, the resulting bounds. On offboard tracks pose improves on neither no pose nor a stranger's pose at three training scales. On onboard-quality tracks, without the leak, it does not beat no pose on held-out scenes and beats a stranger's only where the tracker is wrong, so a benefit there is neither shown nor excluded.

II. Related Work

Trajectory forecasting on WOMD is dominated by anchor- and query-based transformers. We build on MTR [9], which is representative and strong. Adding raw LiDAR features to a WOMD forecaster leaves pedestrian minADE at 0.54 m [10]. Human mesh recovery from LiDAR is maturing [11], with mo-

tion priors completing sparse observations [12], [13]. Intent-focused work on pedestrian datasets [14], [15] targets crossing decisions rather than metric trajectories. Kress et al. [6] report pose-based gains for vulnerable road users from a research vehicle, at close range and with camera-derived pose. Methodologically our closest precedent is Makansi et al. [16], who find by feature attribution that trajectory predictors draw almost nothing from an input the field had assumed was central.

Relation to prior positive results. Pose conditioning has reported real gains, and our results suggest where they come from. Social-Transmotion [3] lowers ADE on the synthetic JTA dataset from 0.99 to 0.89 with ground-truth 3D pose and from 0.40 to 0.36 on JRDB with estimated pose. Social-Pose [4] reports about 25% on JTA ($1.20 \rightarrow 0.90$) and on JRDB. Salzmann et al. [5] report $\approx 10\%$ from onboard-estimated keypoints at ranges up to 7 m, for agents with little position history. A short history is a poor track and 7 m is a well-resolved body, the one condition under which our pre-fit poses beat a stranger's (Sec. IV-E). Le et al. [17] localise the JRDB gain to lower-body 3D keypoints (13% ADE).

The two vehicle-captured settings closest to ours report small effects. Li et al. [18] add 3D keypoints to a predictor on the Waymo perception set, and 4 s minADE moves from 0.249 to 0.239 m in a single run. Social-Pose also evaluates on Pedestrians and Cyclists in Road Traffic, and with ground-truth 3D pose ADE moves from 0.58 to 0.57 m, by 0.01 m or less for all five architectures tested, against the $\approx 25\%$ the same encoder delivers on JTA. Two properties separate the remaining positive settings from an AV stack. The first is range and sensor. JRDB is an indoor and campus robot dataset in which pedestrians are metres away and pose is camera-derived, whereas WOMD pedestrians are observed from 2 to 75 m and often return a few dozen LiDAR points. The second is the baseline. Those predictors consume positions, not a box heading, so body facing is new information there and partly duplicated here. None reports a shuffled-pose arm.

III. Setup

A. Data

We use WOMD [7] with SMPL [19] poses recovered from LiDAR by the Waymo-3DSkelMo pipeline [20], which fits per-frame bodies with LiDAR-HMR [11] and completes them with a NeMF motion prior trained on AMASS [12], [13]. We run its released second version, whose fit adds a term that keeps the joints inside the labelled box. Poses cover the 11 observed frames, and the prediction horizon is 8 s at 10 Hz. In every arm the model sees the map and the pedestrians that carry poses. Vehicles, cyclists and the ego vehicle are not in its input. Every predicted agent carries at least one valid pose frame, and 86.7% of observed frames carry one.

One property of that pipeline bears on Sec. V: when a pedestrian is moving and the raw LiDAR estimate of root rotation deviates from the past-trajectory direction by more than 90°, the pipeline snaps the root to the direction of travel. Body facing is therefore partly trajectory-derived for moving pedestrians. This reinforces the redundancy we report rather than weakening it, and it does not reach the stationary-onset analysis, whose events are all below 0.3 m/s. A second property bears on Sec. IV-E: the motion-prior stage fits each sequence against the box track, so the completed body is not independent of that track. We call the per-frame LiDAR-HMR output that precedes it the pre-fit pose (Fig. 3).

Applying the pipeline to WOMD requires one correction. Its export groups pose clips by object identifier, and WOMD identifiers are unique only within a scene, so nearly every training sequence carries another pedestrian's body. Grouping per scene ($299{,}220$ scenes, 1.81M sequences) cuts the median error between root heading and travel direction from 41.5° to 11.4° and raises the gait–speed correlation from $r{=}0.26$ to 0.58. The uncorrected export fails silently, since training converges and losses look normal. All results use the corrected export, in which the body's root is nearer to its own track's box centre than to any other agent's on 99.0% of posed validation frames.

B. Arms and the content test

Three arms share one architecture (Fig. 2). The no-pose arm receives trajectories and map. The pose arm adds the pedestrian's own SMPL sequence, 144 joint-rotation values per frame with no translation, through a cross-attention encoder with sinusoidal positional encoding. The shuffled arm receives a stranger's sequence, that of another pedestrian assigned once over the whole split by a fixed derangement, with identical parameter count, optimiser and input dimensionality. The root orientation enters in world coordinates while the trajectories are agent-centric, so this arm does not state facing relative to the model's frame. The agent-frame-root encoding does, and its content test is null too ($+0.68\%$, $p{=}0.49$, $n{=}3$, MDE 4.3 pp). A fourth arm, the oracle, replaces the SMPL features with a six-channel encoding of the agent's own future displacement at 3 and 8 s, leaving the branch and its masks untouched. It shows the pathway can carry information at all (Sec. IV-C). Every arm but the no-pose arm, with pose by branch or by early fusion, also has a head for a future pose sequence. WOMD releases LiDAR for the observed second only, so its loss has practically no targets and is inactive in every arm.

Pose versus no-pose asks whether pose improves the forecast, and confounds information with the capacity of the added branch. Pose versus shuffled holds both fixed, isolates content and so attributes the outcome. We call it the content test. Relative differences are $(A - B)/B$, so negative favours A.

Inference is over seeds, the unit the treatment is applied to. Unless flagged, every comparison between trained arms is a paired t-test on per-seed differences.

C. Protocol

Every arm trains for 30 epochs with AdamW (two scenes per GPU with at most 24 predicted pedestrians each, weight decay 0.01, gradient-norm clipping 1.0) under a step schedule that halves the learning rate at epochs 10, 20 and 25, and

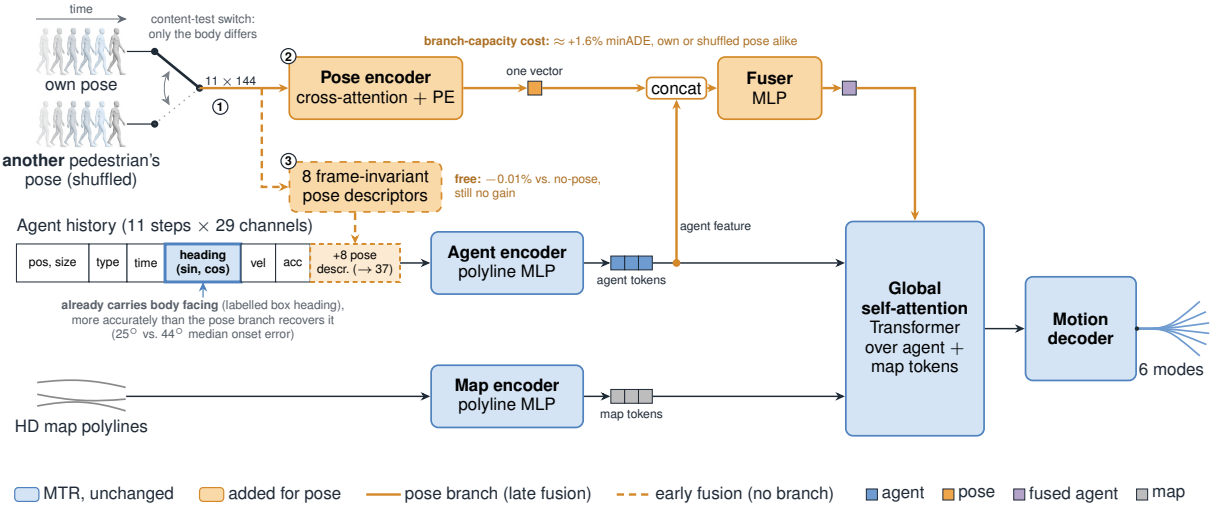


Fig. 2. The two pose arms differ only in whose body enters the model. The content-test switch (1) feeds the predictor either the pedestrian's own SMPL sequence or another pedestrian's, holding parameters, optimiser and input dimensionality fixed. Blue: MTR, unchanged. Orange: added for pose. The agent-history strip shows that the labelled box heading is already an input. Solid orange (2): the pose branch (late fusion). Dashed (3): early fusion (Sec. IV-B).

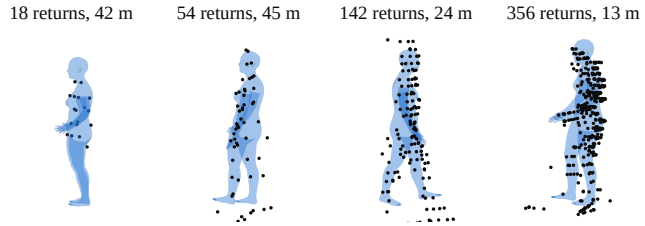


Fig. 3. LiDAR returns (black) and the pre-fit SMPL body estimated from them (blue), at one metric scale. One scored pedestrian per stratum of mean returns over the observed second, cut at 30, 100 and 200: the pedestrian at the stratum median, in its frame closest to that mean. Titles give the frame's returns and range.

TABLE I

POSE AGAINST NO POSE AND AGAINST SHUFFLED ACROSS TRAINING SCALE, VALIDATION FIXED. THE LEARNING RATE IS SELECTED PER SCALE BY A GRID RUN SEPARATELY FOR EVERY ARM, AND ALL ARMS PEAK AT THE SAME VALUE (SEC. III-C). PERCENTAGES ARE DIFFERENCES OF ARM MEANS, AND p-VALUES ARE PAIRED t-TESTS OVER PER-SEED DIFFERENCES. NEGATIVE FAVOURS POSE.

Training scenes	579	5k	25k
Pedestrians	6,348	31,302	142,807
Seeds	5	3	5
minADE (m)			
No-pose	0.4599	0.4176	0.3716
Pose	0.4620	0.4138	0.3778
Shuffled	0.4589	0.4181	0.3777
Pose vs. no-pose	+0.45%	−0.90%	+1.66%*
Content test	+0.66%	−1.01%	+0.04%
p (content)	0.52	0.56	0.93

*p=5×10^{-4}. Shuffled vs. no-pose: +1.62%, p=0.048.

the decoder predicts K=6 modes. The learning rate is chosen per scale from a grid run separately for each arm: 1×10^{-4} at 579 scenes, 5×10^{-5} at 5k and 2.5×10^{-5} at 25k, where every arm peaks at the same interior optimum. Validation is a fixed 5,171 of the 27,156 exported validation scenes. Of their 17,218 predicted pedestrians, the 16,486 with at least one valid future step are scored. Headline numbers are taken at each run's best validation epoch over a fixed evaluation schedule, a selection made on the reported set and identically for every arm, and Sec. IV-B reports what changes under a fixed-epoch rule. Official WOMD mAP needs the full retained prediction set, which our per-epoch evaluation does not store, so we report minADE, minFDE and miss rate at 2 m. minADE averages over a pedestrian's valid future steps and minFDE is taken at the last valid one, and 36% of the scored pedestrians have a full 8 s future. Code, configs, per-seed metrics and the analysis scripts are released with the paper.

IV. RESULTS

A. The null on offboard tracks, and its bound

Table I reports all three arms at three scales. Data alone improves the baseline by 19.2%. Pose does not improve on no pose at any scale and costs 1.66% at 25k (Sec. IV-B). The content test is null at every scale, in minADE, in minFDE (+0.45%, −0.28% and +0.33%, $p \geq 0.61$) and in miss rate at 2 m, where it is +2.2% at 25k (p=0.21) and +4.3% fine-tuned (p=0.19): null, and at 25k with its sign against pose on all three metrics. At 80% power the minimum detectable content effect is 3.5% at 579 scenes (n=5), 7.6% at 5k (n=3) and 2.2% at 25k (n=5). The weight falls on 25k, the most sensitive scale. There the paired seed-level interval on the content difference is $[-1.59\%, +1.70\%]$. The validation scenes contribute less: resampling them with the seeds fixed gives $[-0.4\%, +0.5\%]$. The data thus exclude a content effect beyond roughly 1.7% in either direction.

After pretraining a trajectory-only backbone on full WOMD and fine-tuning at 579 scenes (n=5), the content test is $+0.88\%$ (p=0.18), again with its sign against pose.

B. Two effects that are real, and are not pose

A branch-capacity cost. At 25k both pose arms are worse than no-pose ($+1.66\%$ at $p=5\times10^{-4}$ and $+1.62\%$ at p=0.048) and indistinguishable from each other. An extra input branch costs $\sim$1.6% whether it carries the right body or the wrong one. In both arms every agent's features pass through the branch's fusion layers, which the no-pose arm lacks.

This is measured at each run's best validation epoch. Under a fixed late-epoch rule the sign of a pose-versus-no-pose comparison is not stable (at epoch 30 pose is 4.6% worse at 579 scenes and 5.2% better at 25k, both n.s.), so outside the held-out pass of Sec. IV-E we never compare architecturally different arms at a fixed late epoch. The content test is not significant under either rule, though it is noisier at epoch 30 (-5.9%, -4.7% and -1.7% at the three scales, p=0.16, 0.07 and 0.60). Replacing the 144-d branch with eight frame-invariant descriptors fused into the existing agent channels shows no cost (-0.01% versus no-pose at p=0.99, and -1.67% versus the branch at p=0.12, n=3) without creating a benefit: pose can be carried for free, and still does not pay. Fed an uncorrupted oracle at 25k scenes this pathway converts it on every seed, by -31.2% (p=0.0002) against the branch's -34.5%, so a channel it costs nothing to carry is still a channel it reads.

Tuning dominates the treatment. Across the 25k grid the learning rate moves minADE by up to 16.7% against a content effect of 0.04%, so every arm is tuned separately.

C. The branch is not inert

A null measured through one pathway invites the objection that the pathway is dead, and the shuffled control does not dispose of it: a branch that transmits nothing would also fail to tell its own body from a stranger's.

The oracle arm of Sec. III is the positive control. It is predictive by construction, and it keeps the branch, its parameter count, its input dimensionality and its per-frame validity masks exactly as the pose arm has them, so the oracle is visible only where a real pose would have been.

Over three seeds at 579 scenes the oracle arm changes minADE by -34.8% against no pose ($t(2)=-81.4$, $p=1.5\times10^{-4}$) and by -35.6% against the shuffled arm. The oracle calibrates the pathway: the branch converts information into accuracy at roughly twenty times the magnitude of the bound our content test places on pose. The null is therefore a statement about what SMPL pose contains, not about whether the architecture can use what it is given. That the branch also reads joint rotations is shown by the fitted poses of Sec. IV-E.

D. Where a benefit could still hide

At 25k the content test is also null at the short horizons (-0.6% at 2 s and $+0.1\%$ at 4 s, p=0.46 and 0.90). Across quartiles and top deciles of constant-velocity residual, heading change, speed range and peak acceleration, on the 13,976 agents for which the four are defined, one of 20 tests falls below p=0.05. In the top decile of heading change own pose is 2.0% ahead of a stranger's (p=0.009, 0.19 after Bonferroni correction) and 1.9% behind no pose (p=0.08). The content test is null for standing-then-moving pedestrians, the subpopulation built to favour it (n=1,809 pedestrians, $+0.38\%$, epoch-30 checkpoints on all 27,156 validation scenes, p=0.80 by a bootstrap over seeds and pedestrians).

It is also null where the body is best measured, which is where a pose-quality explanation predicts an effect should appear. Stratifying the 25k content test by LiDAR return count gives -0.37% ($[-1.87, +1.14]$) on the 2,372 agents with at least 200 returns and -0.08% on the 10,902 with at least 30, and stratifying by range gives -0.14% ($[-1.61, +1.33]$) inside 10 m. No stratum shows an effect, at n=5 throughout.

Across the more than forty content tests this study reports outside the onboard-track ladder of Sec. IV-E, three reach $p<0.05$. One is the heading-change decile above, and two favour the stranger: the stationary subpopulation of the heading ablation ($+1.8\%$, n=8, p=0.011) and the stationary subset of the frame-rotation control at $\sigma=30^\circ$ ($+1.3\%$, n=3, p=0.049, Sec. V-B). At that multiplicity two or three such results are expected by chance, and these disagree in direction.

TABLE II

DEGRADED INPUTS AT 25K SCENES, n=3 SEEDS PER ARM FOR THE FRAME ROTATION, WHOSE SEEDS THE CLEAN ROW USES, AND n=5 FOR TRACK NOISE. LOSS IS THE NO-POSE BASELINE AGAINST ITS CLEAN COUNTERPART ON THE SEEDS BOTH SHARE, AND NEGATIVE FAVOURS POSE. THE FRAME ROTATION LEAVES THE CLEAN HEADING READABLE, SO IT IS NOT AN ONBOARD-QUALITY HEADING (SEC. V-B). THE FITTED POSES SAW THE OFFBOARD TRACK, SO THEIR TRACK-NOISE ROWS OVERSTATE WHAT THE BODY ADDS. THE LAST ROWS REPEAT THE 30 CM LEVEL ON PRE-FIT POSES, OVERALL AND BY MEAN LIDAR RETURNS.

Degraded input	Loss	Content	p	vs. no-pose
None	n/a	$+0.41\%$	0.60	$+1.97\%$
Agent frame rotated by a fixed error $\mathcal{N}(0,\sigma^2)$				
$\sigma=15^\circ$	$+5.8\%$	-0.14%	0.90	-0.39%
$\sigma=30^\circ$	$+12.6\%$	-0.11%	0.93	$+1.34\%$
Track, causal tracker on detection noise s, fitted poses				
s=10 cm	$+14.1\%$	-2.08%	0.005	-1.21%
s=30 cm	$+32.3\%$	-5.32%	0.002	-4.06%
Track at s=30 cm, pre-fit poses				
All pedestrians	$+32.3\%$	-1.34%	0.10	$+0.36\%$
<30 returns	n/a	-0.34%	0.72	$+1.82\%$
≥ 200 returns	n/a	-3.47%	0.010	-1.99%

E. An onboard-quality track, and a label leak

Everything above is measured on WOMD's tracks, and those are offboard auto-labels. An offline network refines every box with the whole track, future frames included [7], [8], and the velocity the baseline receives is exactly the backward difference of those refined positions at all 954,751 pedestrian time steps we checked. The observed history is as clean as the labeller could make it. A deployed stack has a causal tracker instead, so we simulate one. Each observed position receives detection noise $\mathcal{N}(0, s^2)$ per axis, drawn once rather

than per epoch, and a forward-only constant-velocity Kalman filter, started at rest at the first observed frame (acceleration noise $1\,\mathrm{m/s^2}$, initial velocity uncertainty $1.5\,\mathrm{m/s}$), turns the detections into the positions and velocities the model sees. The box heading and the future targets are untouched. At prediction time, measured on 675 pedestrians, the tracker is off by $7.5\,\mathrm{cm}$ and $0.23\,\mathrm{m/s}$ in median at $s{=}10\,\mathrm{cm}$, and by 21 cm and $0.46\,\mathrm{m/s}$ at $30\,\mathrm{cm}$, against a median pedestrian speed of $1.1\,\mathrm{m/s}$.

With the pipeline's fitted poses the null breaks (Table II). At $30\,\mathrm{cm}$ own pose beats no pose by 4.06% ($p{=}0.004$) and a stranger's by 5.32% ($p{=}0.002$, 95% CI $[-7.3, -3.4]$), over five seeds. The content effect holds at every fixed late epoch (-5.2 to -5.8% at epochs 26, 28 and 30). It sits where gait would put it: on the 8,757 pedestrians walking at 0.8 to $1.6\,\mathrm{m/s}$ it is -6.8% ($p{=}0.0002$), while on the 3,336 standing at prediction time it is not significant.

Most of it is the label. The fitted poses are not independent of the track. The motion-prior stage [12] is a latent optimisation whose loss predicts the root translation from the local body motion and matches it to the box track, so the fit pulls the gait toward a stride that reproduces the offboard displacement. The fit runs on the observed second alone, so no future frame enters, and the model input holds joint rotations only, so auditing the input does not reveal it. Derived inputs have carried labels before: central-difference velocities gave Trajectron++ the next position [21]. A check without training does. On the 11,588 validation pedestrians with a pose on every observed frame, gradient-boosted trees predict the clean one-second speed from body pose alone, root excluded, with $R^2{=}0.94$ from the fitted pose and 0.44 from the pre-fit one. On moving pedestrians with fewer than 30 LiDAR returns it is 0.78 against less than zero: the fitted body knows a speed the sensor could not have shown it. Added to a regressor on the $30\,\mathrm{cm}$ tracker's own state, the fitted pose removes 71% of the squared speed error that regressor leaves, the pre-fit pose about a fifth and a stranger's 1%.

We therefore train both pose arms at $30\,\mathrm{cm}$ on the pre-fit poses, which see only the points inside the box. Files, agents, tracks and the stranger assigned to each pedestrian are unchanged, and the two shuffled arms agree ($+0.40\%$, $p{=}0.54$). Own pose then does not beat no pose ($+0.36\%$, $p{=}0.54$) and is 1.34% ahead of a stranger's ($p{=}0.10$, CI $[-3.1, +0.4]$). At fixed epochs 26, 28 and 30 the content effect reads -1.4, -2.6 and -1.5%, significant at the second only ($p{=}0.02$). The fit accounts for 3.98 of the 5.32 points ($p{=}0.0001$, between 3.5 and 4.9 on every seed), and on walkers -6.8% becomes -0.8% ($p{=}0.36$).

What survives appears to follow the sensor. Split by mean LiDAR returns over the observed second, the pre-fit content effect is -0.3% below 30 returns, -1.0% and -1.4% in the two middle strata, and -3.47% ($p{=}0.010$, CI $[-5.6, -1.4]$) on the 2,365 pedestrians with at least 200 that every run scores, where own pose also beats no pose (-1.99%, $p{=}0.041$). The top stratum is better than the bottom on every seed, by 3.1 points ($p{=}0.004$), and at every fixed late epoch it reads -3.2 to -4.6%. At those epochs the bottom stratum reads -1.4 to -2.1% (n.s.), so the gap narrows to 1.7 to 2.5 points. The fitted poses show no gradient (-5.3, -5.4, -4.1 and -6.0%, $p{=}0.41$), as a leak that does not depend on the sensor would not. On the offboard track the same stratum is null (Sec. IV-D).

TABLE III

WHERE THE PRE-FIT GAIN SITS. EACH ROW SORTS THE 2,272 PEDESTRIANS WITH AT LEAST 200 RETURNS AND 1 S OF FUTURE BY ONE MEASURE AND CUTS THEM INTO QUARTILES OF 568 (Q1 LOWEST). CELLS GIVE THE GAIN AT $s{=}30$ CM AT EACH RUN'S BEST VALIDATION EPOCH: MINADE WITH THE PEDESTRIAN'S OWN POSE AGAINST NO POSE, IN PERCENT OVER $n{=}5$ SEEDS, NEGATIVE FAVOURING OWN POSE. TRACKER ERROR IS THE DISTANCE OF THE TRACKED STATE FROM THE LABEL AT THE LAST OBSERVED FRAME. TREND IS Q4 MINUS Q1 IN POINTS WITH ITS PAIRED p, AND ASTERISKS MARK $p{<}0.05$, UNCORRECTED.

	Q1	Q2	Q3	Q4	Trend (p)
Tracker error at prediction time					
Velocity	$+0.4$	-0.2	-4.7^*	-3.3^*	-3.7 (0.013)
Position	$+0.1$	-1.2	-1.8	-5.0^*	-5.1 (0.019)
Manoeuvre over the labelled 8 s future					
Constant-velocity residual	-5.0^*	-1.9	-2.2	-1.2	$+3.8$ (0.048)
Heading variation	-3.4^*	-1.3	-2.0	-1.9^*	$+1.5$ (0.21)
Speed range	-5.6^*	-3.3^*	$+0.4$	-1.5	$+4.1$ (0.019)
Peak acceleration	-3.8^*	-2.4^*	-0.9	-1.6	$+2.2$ (0.25)

It follows the tracker's error, not the manoeuvre. Table III splits the 2,272 well-resolved pedestrians with at least 1 s of future two ways and scores own pose against no pose, -2.1% on all of them ($p{=}0.039$). By how wrong the tracker is at prediction time, the gain is absent in the quartile with the smallest velocity error ($+0.4\%$) and -3.3% in the largest, a difference of 3.7 points ($p{=}0.013$, examples in Fig. 4), and $+0.1\%$ against -5.0% over position error (5.1 points, $p{=}0.019$). The velocity trend holds on complete one-second histories alone (4.0 points, $p{=}0.019$). By what the pedestrian goes on to do, the four manoeuvre measures of Sec. IV-D, the gain is largest in the lowest quartile of every measure (-3.4 to -5.6%) and shrinks toward the highest, by 3.8 points over constant-velocity residual ($p{=}0.048$) and 4.1 over speed range ($p{=}0.019$). Against a stranger's pose, which holds the branch fixed, own pose is ahead in all 24 quartiles, with the same trend over tracker error (3.1 and 5.5 points, both $p{=}0.001$) and none over the manoeuvre ($p \geq 0.19$).

Three caveats bound this. The tracker is simulated, with Gaussian detection noise rather than a detector's correlated, occlusion-driven errors, and it never switches or fragments an identity, errors that degrade forecasts on real tracker output [22]. It also never follows a pedestrian for more than a second. On a steadily walking pedestrian the same filter's median velocity error at $30\,\mathrm{cm}$ is $0.33\,\mathrm{m/s}$ after one second and $0.16\,\mathrm{m/s}$ after three, so a longer-lived track leaves pose less to repair. The pre-fit poses are per-frame and unsmoothed, so a prior fitted to the body alone might recover more. The fit used here does not: on 1,789 keypoint-labelled frames of the perception validation set [23], [24] the fitted poses are less accurate than the pre-fit ones (94.8 against 90.7 mm MPJPE, aligned at the hip midpoint), and the loss is at the knees and ankles. Finally, $p{=}0.010$ and $p{=}0.041$ are uncorrected over the eight strata scored at this noise level, and the splits of

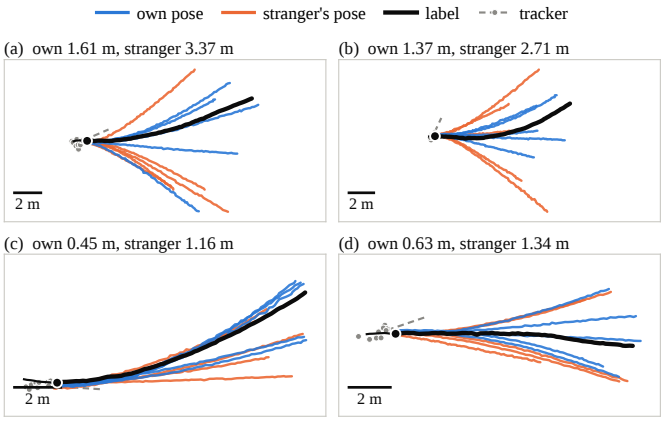


Fig. 4. Four pedestrians on which own pose helps under the 30 cm tracker, drawn in the label frame at prediction time. Each coloured line is the best of six modes of one seed at the final epoch, and titles give the mean minADE. The examples are selected on the effect. Among the 568 well-resolved pedestrians with the largest tracker velocity error, own pose beats a stranger's on all five seeds for 43 and loses on all five for 28 at that epoch, and these are the four largest gains with a complete future. Dots are the tracked history, and the dashed line extends the tracker's velocity over 3 s.

Table III were chosen after that result.

Held-out scenes keep the trend and do not confirm the gain over no pose. We fixed the stratum, the splits and the tests, and scored the fifteen models at their final epoch, the only checkpoint kept, on the 5,648 validation scenes that hold a pedestrian of at least 200 returns and played no part in training, in selecting an epoch or in choosing the stratum, 13,937 such pedestrians in all. Own pose is ahead of a stranger's by 2.1% (p=0.12, CI $[-5.0, +0.8]$), against 3.3% (p=0.045) on the original scenes in the same pass. On the 13,350 with 1 s of future, over the quartiles of the tracker's velocity error, this reads 0.0, −1.4, −1.7 and −4.9%, a trend of 4.9 points (p<0.001). Against no pose the trend is 4.4 points (p=0.017), but the gain is not confirmed: +0.9% (p=0.67, CI $[-4.3, +6.0]$), and −0.1% on the original scenes at the final epoch, where a stranger's pose costs 3.3% (p=0.035). Such comparisons between architectures are unstable at a fixed epoch (Sec. IV-B), and the best-epoch rule behind the −2.1% against no pose in Table III need not be neutral either: from it the pose arms lose 3.1 and 3.3% to the final epoch and the no-pose arm 1.4%. On a causal track own pose is worth more than a stranger's where the tracker is wrong. Carrying a stranger's pose costs about as much, so a gain over no pose is not established.

V. What Limits Pose on Offboard Tracks

A. What the oracle ladder measures

The content test bounds what pose is worth, and the oracle of Sec. IV-C shows the branch can carry information. Neither says what kind of information this pathway turns into accuracy, so we degrade the oracle.

Each level of Table IV feeds the branch the same direction-only oracle, the bearing of the agent's own future displacement at 3 and 8 s, corrupted by a fixed angular error. Each is labelled by the median bearing error actually delivered: 0.6745σ by construction, since the corruption is a fixed per-agent rotation, and measured to within 0.7° of it on 11,328 scored agents. Displacement magnitude is withheld, so the channel carries direction alone.

At 25k scenes every level converts, and the gain stays large: −34.50% with an exact bearing, −19.77% at 5.0°, −15.68% at 10.0° and −13.57% at 15.1°, all at $p \leq 0.0016$ against minimum detectable effects of 1.4 to 2.9 pp, then −10.39% at 24.9° (p=0.003, CI $[-12.9, -7.9]$) and −6.92% at 44.4° (p=0.013, CI $[-10.3, -3.5]$), a fifth of the exact gain (Table IV). The decay is not a single exponential. Two exponentials fit all six levels to 0.3 pp, a fast term of 15.5% with a half-life of 2.1° and a slow term of 19.0% with a half-life of 30°.

The angular axis is not an acceptance threshold. It is tempting to read the bearing error as the accuracy a cue must reach. The ladder refutes that reading itself. The baseline already receives the labelled box heading, as a per-timestep sine and cosine pair for every agent (Fig. 2), and that heading estimates the same future displacement bearing at 3.9° in median, better than the 15.1° channel at every percentile (9.3° against 25.8° at the third quartile, 20.1° against 36.3° at the ninetieth). A channel uniformly less accurate than an input the model already has still buys −13.6%, and the only content it has is the realised future.

The slow term is future-side mode selection. The two terms behave differently in ways that identify them. Per-agent records at each run's own best epoch, paired between arms on 16,486 agents and averaged over seeds, put the slow term at the end of the trajectory: on the 5,888 with a full 8 s future the 15.1° channel keeps 25% of the exact oracle's error reduction at 2 s, 40% at 8 s and 54% at the endpoint. The selection reading predicts more: corrupting the bearing should barely affect which of the six modes is right, and it barely does. The gap between the top-ranked mode and the best of six, which is what the forecaster loses by choosing wrongly, falls from 0.67 m to 0.26 m with an exact oracle, and still to 0.29 m at 15.1°: the corrupted channel closes 92% of the perfect oracle's selection gap while recovering 39% of its minADE. A coarse future bearing says which way the agent went, and only a precise one refines where along that way, which a few degrees of error destroy.

minADE takes the best of six modes whatever their rank, so selection reaches it only through the decoder's scoring step, which keeps six of 64 intention queries: a better ranking changes which six survive. That is also where a pose effect could hide from our headline metric, so we score the ranking metrics directly on the 25k arms. They are null as well: against a stranger's pose, top-1 ADE is −2.07% (p=0.11, MDE 3.7 pp), best-mode rank −0.93% (p=0.22) and endpoint mode spread +0.17% (p=0.53), over the same 16,486 agents.

TABLE IV

THE ORACLE LADDER AT 25K SCENES. A DIRECTION-ONLY ORACLE CORRUPTED BY A FIXED PER-AGENT ANGULAR ERROR, EACH LEVEL LABELLED BY THE MEDIAN BEARING ERROR ACTUALLY DELIVERED. TESTS ARE PAIRED t AGAINST EACH SEED'S OWN NO-POSE RUN, AND NEGATIVE FAVOURS THE ORACLE. MDE IS THE MINIMUM EFFECT A LEVEL COULD DETECT AT 80% POWER. EVERY LEVEL CONVERTS, OUT TO $44.4°$.

Bearing error	n	ΔminADE	p	95% CI	MDE
$0.0°$	3	-34.50%	0.0001	$[-35.6, -33.4]$	1.4
$5.0°$	3	-19.77%	0.0007	$[-22.1, -17.5]$	2.9
$10.0°$	3	-15.68%	0.0009	$[-17.7, -13.6]$	2.6
$15.1°$	3	-13.57%	0.0016	$[-15.9, -11.2]$	2.9
$24.9°$	3	-10.39%	0.0031	$[-12.9, -7.9]$	3.1
$44.4°$	3	-6.92%	0.0127	$[-10.3, -3.5]$	4.2

B. Why body facing does not pay

The oracle converts a bearing read off the realised future. Body facing is not that: it is an observation-time estimate of a quantity the model already measures. Three properties of it, on the population the ladder scores, say what is left for it to contribute.

It is a noisier copy of an existing input. The box heading estimates the future displacement bearing at $3.9°$ ($+3$ s) and $7.1°$ ($+8$ s) in median, while SMPL facing reaches $12.3°$ (n=5,824) and $15.1°$ (n=6,206). Box heading is the better estimate at every percentile we measured, the two differ by $10.0°$ in per-agent median, and both derive from the same returns. The SMPL forward axis is calibrated against the box heading before either is measured.

Most of its accuracy is the trajectory restated. 95.5% of the scored agents move, and for a mover the pipeline snaps root orientation to the direction of travel whenever the raw estimate deviates by more than $90°$ (Sec. III). Movers sit at $11.9°$, while the 262 stationary agents, the only ones whose facing their velocity does not already imply, sit at $41.2°$. The pooled median largely measures the snapping.

Where it would be non-redundant, it is worst. On the one case where velocity carries no directional information (which way a stationary pedestrian steps off), SMPL facing predicts onset direction at roughly twice chance over the 1,705 onset events among the 19,939 stationary pedestrians of the whole validation split: median error $44.2°$, 50.6% within $45°$ (chance $90°$, 25%). That is informative, and not an artifact of our axis fitting: the box heading estimates the same quantity at $25.2°$, and paired on the same events SMPL is $7.3°$ worse in median (Wilcoxon p=2.6×10^{-43}). Nor does pose take over where perception degrades: over those events the gap favours the box heading in all ten testable strata ($p < 10^{-5}$ each) and widens with range, to $+32.9°$ at 41 to 75 m, because both derive from the same returns.

Facing is therefore redundant where it is accurate and inaccurate where it would be non-redundant, and in neither case is it the selection signal the oracle supplies. What survives it is what the content test measures directly: $+0.04\%$ at 25k scenes, bounded to $\pm1.7\%$.

Removing the redundant input does not rescue it. Redundancy says why little is left for facing to add, but it is not by itself a mechanism that blocks conversion. We therefore delete the box-heading channel, retrain all arms and repeat the content test at 579 scenes and eight seeds. No benefit appears. The difference in content effect between the two conditions is $+2.9$ pp on the 13,006 moving agents, the wrong sign for a benefit that redundancy had been masking, and -0.6 pp on the stationary ones (p=0.65). Deleting the channel costs the no-pose baseline only 1.5% on movers, because velocity stays in the input, so there is little room for facing to fill.

A frame-rotation control, not a degraded heading. The heading the baseline receives is an offboard label too (Sec. IV-E), so it may be better than a deployed tracker's, and facing might pay only where it is worse. Deleting the channel cannot test that: MTR's agent-centric frame, and the intention anchors defined in it, stay aligned to the clean heading. We rotate that frame by a fixed per-agent error $\mathcal{N}(0, \sigma^2)$, which frame, anchors and map inherit, and repeat the content test at 25k scenes and three seeds (Table II). The no-pose baseline loses 5.8% at σ=$15°$ and 12.6% at $30°$, and pose recovers none of it: -0.14% (p=0.90) and -0.11% (p=0.93) against shuffled. This does not test a degraded heading. The per-timestep heading channel stays the clean heading expressed in the rotated frame, so the model can read the added error, and the pose root stays in world coordinates, so the body cannot state the correction. Whether facing pays under an onboard-quality heading is untested here. The clean condition reproduces the main results to within 0.4%.

C. A general condition

Two results locate where pose could have helped. At 2 s, minADE is 0.097 m at 25k scenes, and a perfect future-direction oracle through the same branch removes only 1.4 cm of it (-14.7%): in absolute terms there is almost nothing for any cue to win. At long horizons the residual is dominated by destination and intent, which body configuration does not disclose.

In general form, and offered as a hypothesis from one modality on one dataset rather than a law: *an auxiliary modality improves a forecast only where the baseline's own residual leaves room in absolute terms, where that residual is not dominated by intent the modality cannot observe, and where the modality carries information about the cue that the model's existing inputs do not already carry.* The third clause is not a comparison of marginal accuracies: Sec. V-A converts a channel less accurate at every percentile than an input the model already has. What disqualifies facing is that it is a noisier estimate, from the same returns, of what an existing input already gives (Sec. V-B). Sec. IV-E tests the clause from the other side, and shows it has a trap. A cue whose pipeline was fitted to the input it supplements carries that input, and looks informative exactly when the input is degraded. Redundancy is a property of a modality together with the inputs beside it, and an offboard-labelled benchmark sets those inputs at offboard quality.

VI. Practice and Limitations

For an AV stack, three consequences follow. First, before integrating an auxiliary modality, measure how much of its cue it explains beyond the inputs the model already receives, take those inputs at the quality a vehicle will have, and ask what the cue's own pipeline was fitted to. None of this costs training: our fitted poses predict the labelled speed with $R^2{=}0.94$, which flags the leak before any model is trained. Second, an oracle ladder is worth its few runs as a positive control and an upper bound on the pathway, if its MDE is below the effect one would act on (Sec. V-A). Third, the cost of an extra branch ($\sim$1.6% here, at the best-epoch operating point) is easily mistaken for a modality effect in either direction. A shuffled-pose control separates the two, and early fusion avoids the cost while still reading the channel.

Limitations. We test one forecaster, one dataset and one pose estimator, the model sees pedestrians and map only, so pose cannot act through interaction with vehicles and the baseline is not comparable with published WOMD results, and we do not model pose–pose interaction. The largest scale we train is 25k of the 299,220 exported scenes, a compute limit, and the ladders run at $n{=}3$ per level, five for track noise. We did not test a degraded heading (Sec. V-B). The onboard track is simulated (Sec. IV-E). The pre-fit poses are still estimated from points cropped by offboard boxes, the pre-fit control covers the 30 cm level only, and we have not run either through early fusion. The null on offboard tracks is measured with fitted poses. Pre-fit poses were not trained on clean tracks, and the nearest evidence is the quartile of Table III in which the tracker is nearly right, where they gain nothing over no pose. WOMD ships camera data only as quantised tokens, so a camera-based pose or gaze estimate cannot be tested at this scale.

VII. Conclusion

Body pose, recovered from LiDAR, does not improve pedestrian trajectory forecasting on WOMD's offboard tracks. The null is controlled against a stranger's pose and holds across scale, representation and integration, and an oracle through the same branch rules out a dead pathway. Under a simulated onboard tracker own pose beats no pose by 4.1% and a stranger's by 5.3%, and three quarters of the latter is the offboard label, carried in by a pose pipeline that fits the gait to the track. Against a stranger's, poses taken before that fit gain 1.3% overall ($p{=}0.10$) and 3.5% on the best-resolved bodies. On held-out scenes that is 2.1% ($p{=}0.12$), it grows with the tracker's error, and a gain over no pose is not confirmed. The results fit one reading: a cue pays only for what it carries beyond the model's existing inputs, an offboard track leaves the body nothing to add, and a cue fitted to a label carries the label.

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